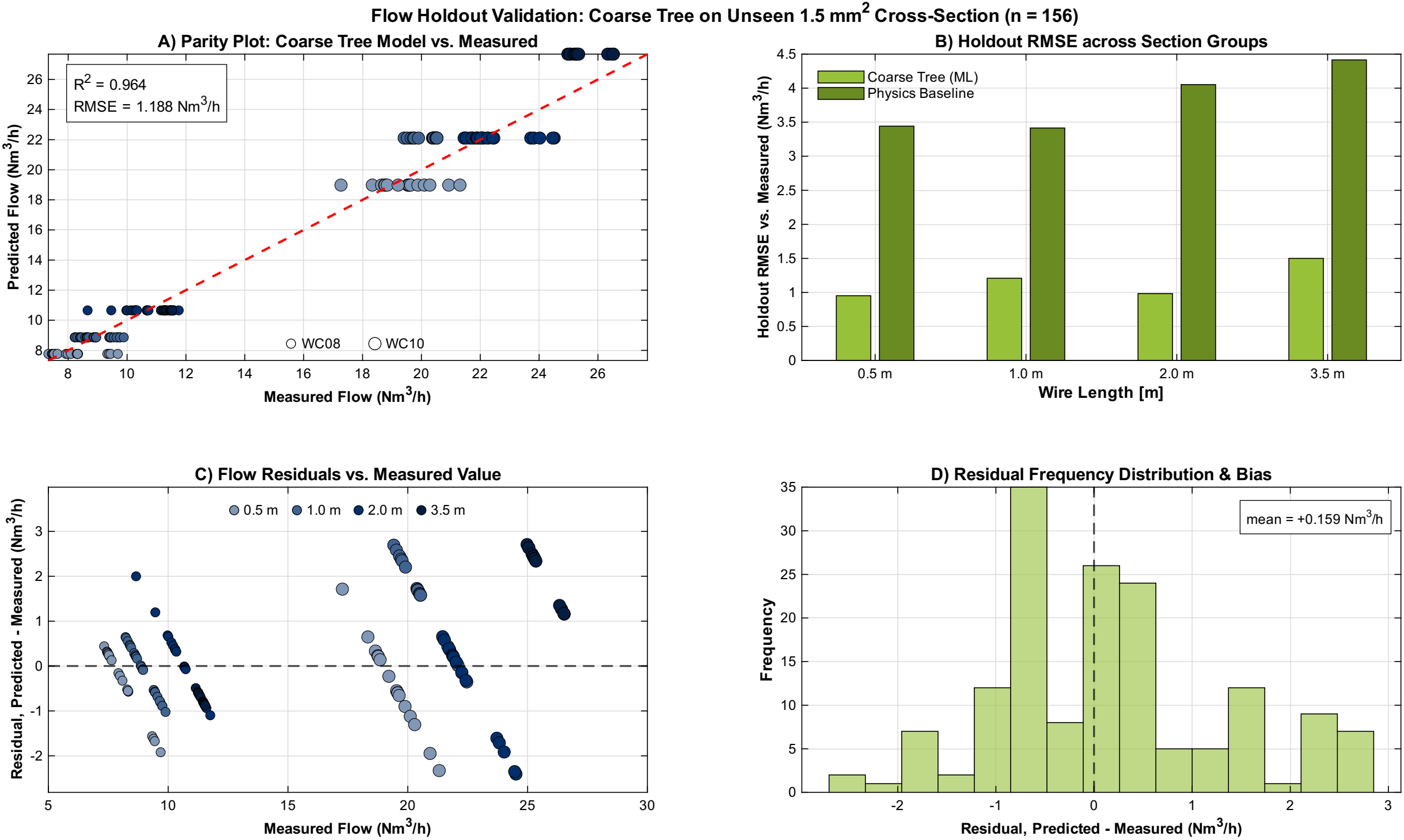

Flow Holdout Validation: Coarse Tree on Unseen 1.5 mm² Cross-Section (n = 156)
A) Parity Plot: Coarse Tree Model vs. Measured
R² = 0.964
RMSE = 1.188 Nm³/h
WC08
WC10
Predicted Flow (Nm³/h)
Measured Flow (Nm³/h)
B) Holdout RMSE across Section Groups
Coarse Tree (ML)
Physics Baseline
Holdout RMSE vs. Measured (Nm³/h)
0.5 m
1.0 m
2.0 m
3.5 m
Wire Length [m]
C) Flow Residuals vs. Measured Value
0.5 m
1.0 m
2.0 m
3.5 m
Residual, Predicted - Measured (Nm³/h)
Measured Flow (Nm³/h)
D) Residual Frequency Distribution & Bias
mean = +0.159 Nm³/h
Frequency
Residual, Predicted - Measured (Nm³/h)